

A 3-REGULAR COUNTEREXAMPLE TO THE BILU–LINIAL SIGNING CONJECTURE

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ABSTRACT. We construct a finite connected simple cubic graph F such that every signing of its edges yields a signed adjacency matrix with an eigenvalue outside $[-2\sqrt{2}, 2\sqrt{2}]$. This disproves the Bilu–Linial signing conjecture for general regular graphs. The graph F is not Ramanujan, and the conjecture restricted to Ramanujan base graphs remains open.

1. INTRODUCTION

1.1. Problem setup. Throughout this paper, all graphs are finite, simple, and undirected, unless explicitly stated otherwise. A graph is called d -regular if every vertex has exactly d neighbours, and a 3-regular graph is called *cubic*. For a graph G , its vertex and edge sets are denoted by $V(G)$ and $E(G)$. A *signing* is a function $\sigma : E(G) \rightarrow \{-1, 1\}$. The signed adjacency matrix $A_\sigma(G)$ is indexed by $V(G)$ and is defined by

$$(A_\sigma(G))_{uv} = \begin{cases} \sigma(uv), & \{u, v\} \in E(G), \\ 0, & \{u, v\} \notin E(G). \end{cases}$$

Here uv denotes the unordered edge $\{u, v\}$; all diagonal entries are zero. We write $A(G)$ for the unsigned adjacency matrix, obtained by taking all edge signs to be $+1$. For a real symmetric matrix K , its Euclidean operator norm is

$$\|K\| = \max_{x \neq 0} \frac{\|Kx\|_2}{\|x\|_2} = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } K\}, \quad \|x\|_2^2 = \sum_i x_i^2.$$

The vector in the maximum ranges over the real coordinate space on which K acts, and the sum ranges over all its coordinates. Expanding in an orthonormal eigenbasis also gives

$$\|K\| = \max_{x \neq 0} \frac{|x^\top K x|}{\|x\|_2^2},$$

where $^\top$ denotes transpose. The quotient without the absolute value is the *Rayleigh quotient* of x with respect to K .

Bilu and Linial [1, 2] asked whether every d -regular graph admits an assignment of signs $+1$ or -1 to its edges such that all eigenvalues of the resulting signed adjacency matrix lie in the interval $[-2\sqrt{d-1}, 2\sqrt{d-1}]$. They formulated the following conjecture (see also Conjectures 6.9 and 6.10 in [5]).

Conjecture 1.1 (Bilu–Linial [2, Conjecture 3.1]). *For every integer $d \geq 2$ and every finite simple d -regular graph G , there exists a signing $\sigma : E(G) \rightarrow \{-1, 1\}$ such that*

$$(1) \quad \|A_\sigma(G)\| \leq 2\sqrt{d-1}.$$

The signing question arises naturally from graph coverings. The *2-lift* associated with a signing replaces each vertex u by two vertices, denoted by u_+ and u_- . Each positive edge uv is replaced by the two edges u_+v_+ and u_-v_- , while each negative edge uv is replaced by the two edges u_+v_- and u_-v_+ . Its adjacency eigenvalues, counted with multiplicity, are those of $A(G)$ together

with those of $A_\sigma(G)$ [2, Lemma 3.1]. Thus, (1) requires all new eigenvalues to lie in the interval $[-2\sqrt{d-1}, 2\sqrt{d-1}]$. We refer to G as the *base graph*.

For a connected d -regular graph, the eigenvalue d , and also $-d$ when the graph is bipartite, are called trivial. A d -regular graph is called *Ramanujan* if all of its nontrivial eigenvalues lie in the interval $[-2\sqrt{d-1}, 2\sqrt{d-1}]$ [9]. This definition clarifies the role of (1): if the base graph is Ramanujan, the bound ensures that all new eigenvalues lie in the Ramanujan interval, so the lift is also Ramanujan. For a comprehensive account of the spectral properties, constructions, and applications of expander graphs, see the survey [5].

1.2. Our contribution. We construct a 3-regular graph such that, for every edge signing, the resulting signed adjacency matrix has at least one eigenvalue outside the interval $[-2\sqrt{2}, 2\sqrt{2}]$. This provides a counterexample to the Bilu–Linial conjecture.

Theorem 1.1. *There exists a finite, connected, simple, 3-regular graph F such that every signing $\sigma : E(F) \rightarrow \{-1, 1\}$ satisfies*

$$\|A_\sigma(F)\| > 2\sqrt{2}.$$

We describe the construction of F in Section 2 and prove in the subsequent sections that it satisfies the conclusion of Theorem 1.1.

1.3. Related work. Marcus, Spielman, and Srivastava [10, Theorem 5.3] proved that every d -regular graph, with $d \geq 2$, admits a signing whose signed adjacency eigenvalues are all at most $2\sqrt{d-1}$. For bipartite graphs, the signed adjacency spectrum is symmetric about zero, so this result establishes the Bilu–Linial conjecture in the bipartite case. Their proof introduced the method of interlacing families of polynomials, which they subsequently developed to resolve the Kadison–Singer problem [11]. Further developments extended this approach to higher-rank versions of Weaver’s KS_r conjecture, with improved quantitative bounds obtained by Xu, Xu, and Zhu [14].

Bilu and Linial [2] proved that every graph of maximum degree $d \geq 2$ admits a signing whose signed adjacency matrix has spectral radius $O(\sqrt{d(\log d)^3})$. For d -regular graphs, Ravichandran and Srivastava [12] improved this bound to $2\sqrt{2(d-1)}$ using interlacing families of mixed determinantal polynomials. See also [7] for an alternative proof of this bound via a graph-theoretic reduction to the bipartite signing theorem. Lin and Zhou [8] further improved this bound, proving that every finite d -regular graph G , with $d \geq 2$, admits a signing σ such that $\|A_\sigma(G)\| < \frac{3+\sqrt{5}}{2}\sqrt{d-1}$.

The covering problem extends naturally to ℓ -lifts. For an integer $\ell \geq 2$, an ℓ -lift replaces each vertex by ℓ copies and each edge by a perfect matching between the corresponding sets of copies. Its spectrum contains that of the base graph; the remaining eigenvalues, counted with multiplicity, are called the new eigenvalues. Hall, Puder, and Sawin [4] proved that, for every integer $\ell \geq 2$, every connected d -regular graph with $d \geq 3$ admits an ℓ -lift whose new eigenvalues are at most $2\sqrt{d-1}$. For a bipartite base graph, spectral symmetry gives the corresponding lower bound. Thus every connected bipartite Ramanujan graph admits a Ramanujan ℓ -lift for every integer $\ell \geq 2$.

1.4. Organization. Section 2 describes the construction of the cubic graph F . Section 3 states the auxiliary lemmas used in the main proof, with proofs provided in Sections 5 and 6. Section 4 uses these lemmas to prove Theorem 1.1. Section 7 shows that F is not Ramanujan. Thus our counterexample disproves the conjecture for general regular graphs but leaves its restriction to Ramanujan base graphs unresolved. The final section presents several open questions for further research.

2. THE GRAPH AND ITS CONSTRUCTION

A *rooted graph* is a graph with a distinguished vertex, called its *root*. Our construction proceeds in three stages. We first define a rooted graph H_{62} , which we call the *seed*. We then join copies of the seed to form the *core* J and complete this core to obtain the 3-regular counterexample F .

Throughout the construction, all graph copies are taken to be pairwise vertex-disjoint before the specified connecting edges are added. In particular, repeated occurrences of the same graph in a recursive formula denote distinct copies.

2.1. The rooted seed H_{62} . For each nonnegative integer h , let T_h be the complete rooted binary tree of height h . The level of a vertex is its distance from the root, measured in edges. Thus, the root is at level zero, each vertex at a level less than h has exactly two children, and the vertices at level h have no children. In particular, T_0 consists of a single vertex. To specify the construction unambiguously, we label the two children of each branching vertex as its first and second children.

We construct the rooted graph H_h as follows. Start with a triangle on vertices v_0, v_1, v_2 , add a new vertex o , and join o to v_0 . Next, take three vertex-disjoint copies of T_h , also disjoint from these four vertices, and join their roots to v_1, v_2 , and o , respectively, by one edge each. We designate o as the root of H_h .

The root o has degree two. The vertices at level h in the three attached trees have degree one in H_h and are precisely its *leaves*, of which there are $3 \cdot 2^h$. Every other vertex has degree three. We use H_{62} as the seed for the counterexample; its structure is illustrated schematically in Figure 1(a).

2.2. Joining copies of the seed to form the core J . For two vertex-disjoint rooted graphs X and Y , let $B(X, Y)$ be the rooted graph obtained by adding a new vertex, joining it to the root of each graph X, Y by an edge, and taking this new vertex as the root. Starting from the seed H_{62} , define recursively

$$(2) \quad Q_0 = H_{62}, \quad Q_{j+1} = B(Q_j, Q_j) \quad (0 \leq j < 61),$$

where the two occurrences of Q_j denote vertex-disjoint copies. The root of each Q_j has degree two. At each recursive step, the roots of the two copies acquire one additional edge and therefore have degree three in Q_{j+1} .

To form the *core* J , take three vertex-disjoint copies of Q_{61} , add a new vertex z , and join z to the root of each copy. The vertex z and the three roots then all have degree three. Consequently, every vertex of J has degree three except the leaves of the binary trees within the seed copies, which retain degree one. Figure 1(b) illustrates the recursive step $Q_{j+1} = B(Q_j, Q_j)$, and Figure 1(c) shows the structure of the core J .

2.3. Completing the core to obtain the 3-regular counterexample F . We complete the core J by applying the following operation separately to each copy of the seed H_{62} . First, order the leaves by listing those in the tree attached to v_1 , followed by those in the tree attached to v_2 , and then those in the tree attached to o . Within each tree, order the leaves lexicographically according to the sequences of first and second child choices along their paths from the root, with the first child preceding the second. Partition the resulting list into consecutive triples; this is possible because each seed has $3 \cdot 2^{62}$ leaves.

For each triple, introduce two new vertices and join each of them to all three leaves in that triple. All newly introduced vertices are distinct, and no other edges are added. In particular, the two vertices introduced for a given triple are not adjacent. Let F denote the graph obtained from J after this operation has been performed in every seed copy.

Each leaf of J gains exactly two incident edges and therefore has degree three in F . Each new vertex also has degree three, while the degrees of all other vertices remain unchanged. Thus, F is 3-regular. Since the new vertices are distinct and each is joined to three distinct leaves, the construction introduces neither loops nor multiple edges. Moreover, F is connected because J is connected and every new vertex is adjacent to vertices of J . Hence F is a finite, connected, simple, 3-regular graph. Finally, J is an induced subgraph of F , since the completion adds no edges between vertices of J . Figure 1 summarizes the construction.

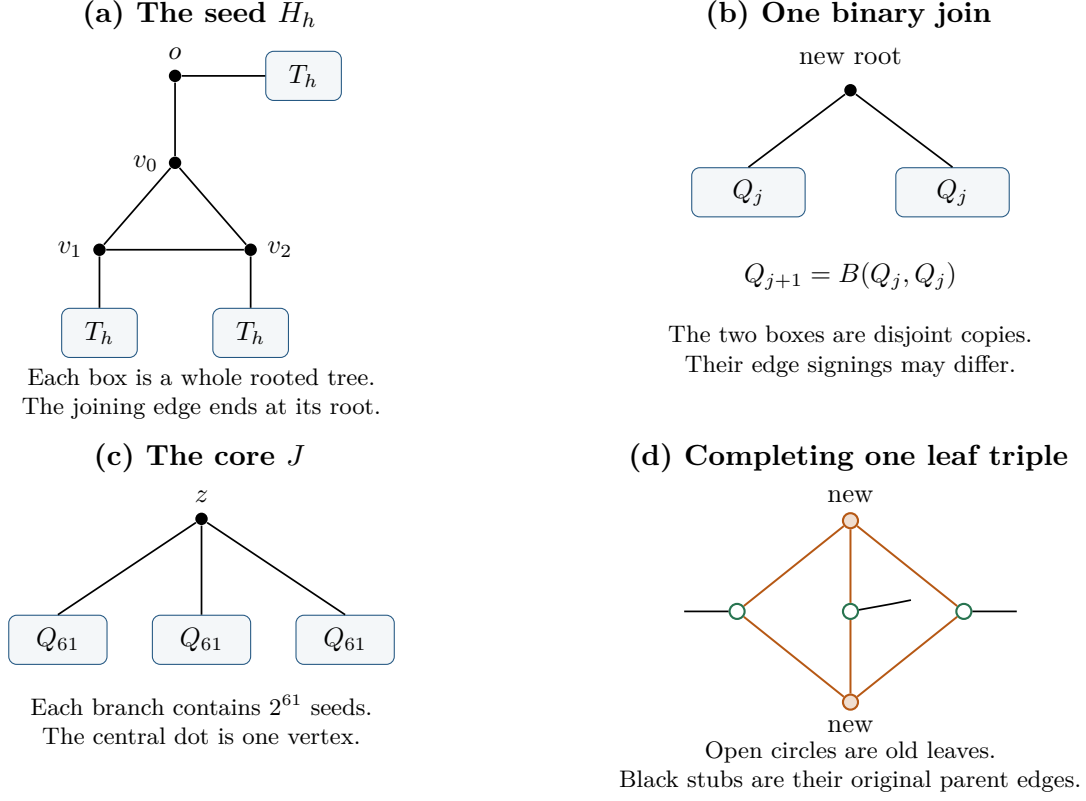


FIGURE 1. The exact construction, with repeated subgraphs compressed into boxes. In (a), set $h = 62$; in (b), perform the joins for $j = 0, \dots, 60$. After (c), carry out (d) separately in each seed, using the leaf ordering specified in the text. Orange vertices and edges are added only at the completion step.

2.4. A small example illustrating the construction. To illustrate the construction with every vertex shown explicitly, Figure 2 presents a smaller graph obtained by joining the roots of three copies of H_0 to a central vertex and then applying the same completion rule. The resulting graph is 3-regular, with 28 vertices and 42 edges. This example serves only to illustrate the construction; we make no claim that it satisfies the spectral conclusion of Theorem 1.1. To illustrate the construction with every vertex shown explicitly, Figure 2 presents a smaller graph obtained by joining the roots of three copies of H_0 to a central vertex and then applying the same completion rule. The resulting graph is 3-regular, with 28 vertices and 42 edges. It serves only to illustrate the construction and we make no claim that it satisfies the spectral conclusion of Theorem 1.1.

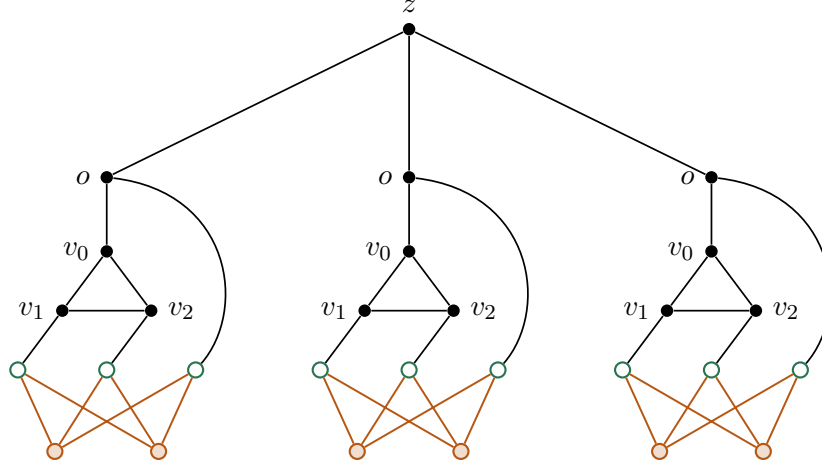


FIGURE 2. A 28-vertex illustration obtained by reducing the tree height to zero and omitting the binary joins. Black dots are seed vertices or the centre, green open circles are old leaves, and orange circles are the six completion vertices. All vertices and edges are shown. Edge crossings without a circle are not vertices.

3. DEFINITIONS AND LEMMAS FOR THE MAIN PROOF

3.1. Schur complements. Schur complements play a key role in our proof. We recall their definition and the standard properties needed below; see [3, Appendix A.5.5, p. 650].

Let ℓ_1, ℓ_2 be positive integers, and let

$$\mathcal{M} = \begin{pmatrix} D & W \\ W^\top & \Gamma \end{pmatrix} \in \mathbb{R}^{(\ell_1+\ell_2) \times (\ell_1+\ell_2)}$$

be a real symmetric matrix, where $D \in \mathbb{R}^{\ell_1 \times \ell_1}$, $\Gamma \in \mathbb{R}^{\ell_2 \times \ell_2}$, and $W \in \mathbb{R}^{\ell_1 \times \ell_2}$. If D is invertible, the matrix

$$\Gamma - W^\top D^{-1} W \in \mathbb{R}^{\ell_2 \times \ell_2}$$

is called the *Schur complement of D in $\mathcal{M}$* . Similarly, if Γ is invertible, the *Schur complement of Γ in $\mathcal{M}$* is $D - W\Gamma^{-1}W^\top \in \mathbb{R}^{\ell_1 \times \ell_1}$.

The Schur complement of D arises naturally when eliminating the variables corresponding to the block D . Indeed, consider the system

$$D\xi + W\eta = 0, \quad W^\top \xi + \Gamma\eta = b,$$

where $\xi \in \mathbb{R}^{\ell_1}$ and $\eta, b \in \mathbb{R}^{\ell_2}$. Since D is invertible, the first equation gives $\xi = -D^{-1}W\eta$. Substituting this expression into the second equation yields the reduced system

$$(\Gamma - W^\top D^{-1} W)\eta = b.$$

The following lemma is standard and we include a proof for completeness. For a real symmetric matrix $K \in \mathbb{R}^{n \times n}$, the notation $K \succeq 0$ means $x^\top K x \geq 0$ for every real vector $x \in \mathbb{R}^n$, and $K \succ 0$ means strict inequality for every nonzero $x \in \mathbb{R}^n$.

Lemma 3.1. *With the notation above, assume in addition that $D \succ 0$. Then $\mathcal{M} \succeq 0$ if and only if $\Gamma - W^\top D^{-1} W \succeq 0$. The analogous equivalence holds with $\succ 0$ throughout. If $\mathcal{M} \succ 0$, the lower-right block of $\mathcal{M}^{-1}$ is $(\Gamma - W^\top D^{-1} W)^{-1}$.*

Proof. We first prove the two positivity equivalences and then identify the lower-right block of the inverse.

Since $D \succ 0$, the matrix D is invertible. For $\xi \in \mathbb{R}^{\ell_1}$ and $\eta \in \mathbb{R}^{\ell_2}$, completing the square gives

$$\begin{aligned} \begin{pmatrix} \xi \\ \eta \end{pmatrix}^\top \mathcal{M} \begin{pmatrix} \xi \\ \eta \end{pmatrix} &= \xi^\top D \xi + 2\xi^\top W \eta + \eta^\top \Gamma \eta \\ &= (\xi + D^{-1}W\eta)^\top D (\xi + D^{-1}W\eta) \\ &\quad + \eta^\top (\Gamma - W^\top D^{-1}W) \eta. \end{aligned}$$

The first term in the last expression is nonnegative and vanishes exactly when $\xi = -D^{-1}W\eta$.

Suppose first that $\mathcal{M} \succeq 0$. Choosing $\xi = -D^{-1}W\eta$ in this identity gives

$$\eta^\top (\Gamma - W^\top D^{-1}W) \eta \geq 0 \quad \text{for every } \eta \in \mathbb{R}^{\ell_2}.$$

Thus the Schur complement is positive semidefinite. If $\mathcal{M} \succ 0$, the same choice gives strict inequality whenever $\eta \neq 0$, because the combined vector (ξ, η) is then nonzero. Hence the Schur complement is positive definite.

Conversely, if the Schur complement is positive semidefinite, both terms in the completed-square expression are nonnegative, so $\mathcal{M} \succeq 0$. If the Schur complement is positive definite, consider any nonzero vector (ξ, η) . When $\eta \neq 0$, the second term is strictly positive. When $\eta = 0$, we have $\xi \neq 0$, and the first term is $\xi^\top D \xi > 0$. Therefore $\mathcal{M} \succ 0$.

It remains to prove the inverse formula. Assume that $\mathcal{M} \succ 0$, so both $\mathcal{M}$ and its Schur complement are invertible. For an arbitrary $\zeta \in \mathbb{R}^{\ell_2}$, write

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = \mathcal{M}^{-1} \begin{pmatrix} 0 \\ \zeta \end{pmatrix},$$

where the zero vector belongs to $\mathbb{R}^{\ell_1}$. Equivalently,

$$D\xi + W\eta = 0, \quad W^\top \xi + \Gamma\eta = \zeta.$$

The first equation gives $\xi = -D^{-1}W\eta$. Substituting into the second yields

$$(\Gamma - W^\top D^{-1}W)\eta = \zeta, \quad \eta = (\Gamma - W^\top D^{-1}W)^{-1}\zeta.$$

By definition, the lower-right block of $\mathcal{M}^{-1}$ maps ζ to η . Since ζ was arbitrary, this block is precisely $(\Gamma - W^\top D^{-1}W)^{-1}$. $\square$

3.2. Root responses and spectral bounds. Throughout the rest of the paper, put $r := 2\sqrt{2}$. For a graph X , let $V(X)$ and $E(X)$ denote its vertex set and edge set, respectively. Write $\mathbb{R}^{V(X)}$ for the space of real-valued functions on $V(X)$, and let I_X denote the identity matrix on this space. In particular, the spectral theorem gives

$$(3) \quad \|A_\sigma(X)\| \leq r \quad \text{if and only if} \quad rI_X - A_\sigma(X) \succeq 0 \quad \text{and} \quad rI_X + A_\sigma(X) \succeq 0.$$

If X is rooted at o_X and both matrices in (3) are positive definite, define

$$(4) \quad \begin{aligned} g_\varepsilon(X, \sigma) &= e_{o_X}^\top (rI_X - \varepsilon A_\sigma(X))^{-1} e_{o_X}, \quad \varepsilon \in \{-1, 1\}, \\ s(X, \sigma) &= g_1(X, \sigma) + g_{-1}(X, \sigma). \end{aligned}$$

The vector e_{o_X} is one at o_X and zero at every other vertex. Each response is strictly positive, since the inverse of a positive definite matrix is positive definite. We call g_ε the *root response* on side ε . Both responses in $s(X, \sigma)$ use the same signing σ . When a signing is fixed on a larger graph, its restriction to the edges of any subgraph is also denoted by σ . An occurrence of Q_j always means the induced copy with its root specified by the construction.

3.3. Auxiliary lemmas. The proof of the main result uses the following three lemmas. Lemmas 3.2 and 3.3 are proved in Section 5, and Lemma 3.4 is proved in Section 6.

Lemma 3.2. *Let σ be a signing of J such that $\|A_\sigma(J)\| \leq r = 2\sqrt{2}$. For each $j \in \{0, \dots, 61\}$ and each rooted copy X of Q_j appearing in the construction of J , we have*

$$rI_X - \varepsilon A_\sigma(X) \succ 0 \quad \text{for every } \varepsilon \in \{-1, 1\},$$

where σ on X denotes the restriction of the signing to $E(X)$. In particular, both root responses $g_1(X, \sigma)$ and $g_{-1}(X, \sigma)$, and hence their sum $s(X, \sigma)$, are well defined by (4), even when $\|A_\sigma(J)\| = r$.

Lemma 3.3. *Let X and Y be vertex-disjoint rooted graphs, and let $Z = B(X, Y)$ be the rooted graph obtained by adding a new vertex and joining this vertex by an edge to each of the roots of X and Y . The new vertex is designated as the root of Z . Suppose that a signing σ of Z satisfies*

$$rI_Z - \varepsilon A_\sigma(Z) \succ 0 \quad \text{for each } \varepsilon \in \{-1, 1\}.$$

Use the same notation σ for its restrictions to X and Y . Then the root responses of X , Y , and Z defined in (4) are well defined. Moreover, for each $\varepsilon \in \{-1, 1\}$,

$$p_\varepsilon := p_\varepsilon(X, Y, \sigma) := r - g_\varepsilon(X, \sigma) - g_\varepsilon(Y, \sigma) > 0,$$

and

$$(5) \quad \begin{aligned} g_\varepsilon(Z, \sigma) &= \frac{1}{p_\varepsilon}, \quad \varepsilon \in \{-1, 1\}, \\ s(Z, \sigma) &\geq \frac{4}{2r - s(X, \sigma) - s(Y, \sigma)}. \end{aligned}$$

The denominator in the second inequality equals $p_1 + p_{-1}$ and is therefore strictly positive.

Lemma 3.4. *For every signing σ of H_{62} , $rI_{H_{62}} - \varepsilon A_\sigma(H_{62}) \succ 0$ for $\varepsilon \in \{-1, 1\}$, and*

$$(6) \quad \frac{s(H_{62}, \sigma)}{\sqrt{2}} = \frac{140300416}{138131009} > \frac{65}{64}.$$

4. PROOF OF THEOREM 1.1

We first outline the main ideas of the proof. The proof proceeds by contradiction on the induced core J . Assuming that a signing satisfies $\|A_\sigma(J)\| \leq r := 2\sqrt{2}$, Lemma 3.2 guarantees that all branch matrices remain positive definite even when the assumed spectral bound is attained with equality. Hence these matrices are invertible, and their root responses $g_\varepsilon, \varepsilon \in \{-1, 1\}$ and the corresponding Schur complements are well defined. Lemma 3.4 provides an initial lower bound for the sum of the two root responses of each seed, and Lemma 3.3 propagates this bound through the binary joins. After 61 levels, each of the three branches attached to the centre has response sum greater than $4\sqrt{2}/3$, so the sum of their response sums exceeds $4\sqrt{2}$. However, taking Schur complements at the centre in $rI_J - A_\sigma(J)$ and $rI_J + A_\sigma(J)$ forces the total of these three response sums to be at most $2r = 4\sqrt{2}$, a contradiction. Since J is an induced subgraph of F , the resulting strict norm bound also holds for every signing of F .

Proof of Theorem 1.1. By the construction in Section 2, the graph F is finite, connected, simple, and cubic. We now prove that $\|A_\sigma(F)\| > r$ for every signing σ of F .

It suffices to consider J . Since J is an induced subgraph of F , every signing σ of F restricts to a signing of J , and $A_\sigma(J)$ is a principal submatrix of $A_\sigma(F)$. Hence

$$\|A_\sigma(J)\| \leq \|A_\sigma(F)\|.$$

It therefore suffices to prove that every signing of J has norm strictly greater than r . Suppose, to the contrary, that a signing σ satisfies $\|A_\sigma(J)\| \leq r$. By Lemma 3.2, both root responses $g_1(X, \sigma)$

and $g_{-1}(X, \sigma)$ are defined for every induced copy X of Q_j in J , with $0 \leq j \leq 61$, even when $\|A_\sigma(J)\| = r$.

For this proof, define

$$c_j = \sqrt{2} \left(1 + \frac{1}{64-j} \right), \quad 0 \leq j \leq 61.$$

We claim that, for each $0 \leq j \leq 61$ and every copy X of Q_j appearing in the construction of J , equipped with the restriction of σ to its edges,

$$(7) \quad s(X, \sigma) > c_j.$$

Here, $s(X, \sigma)$ is defined in (4). We first complete the main argument using this claim and prove the claim afterward.

Let X_1, X_2, X_3 be the three copies of Q_{61} whose roots are joined to the central vertex z . Fix $\varepsilon \in \{-1, 1\}$, and write

$$D_i = rI_{X_i} - \varepsilon A_\sigma(X_i), \quad i = 1, 2, 3.$$

Each D_i is positive definite by Lemma 3.2. Order the vertices of J by listing the vertices of X_1, X_2, X_3 first and z last. Since there are no edges between these three branches, we have

$$rI_J - \varepsilon A_\sigma(J) = \begin{pmatrix} D_1 & 0 & 0 & b_1 \\ 0 & D_2 & 0 & b_2 \\ 0 & 0 & D_3 & b_3 \\ b_1^\top & b_2^\top & b_3^\top & r \end{pmatrix} \succeq 0,$$

where

$$b_i = -\varepsilon \sigma(z o_{X_i}) e_{o_{X_i}}$$

records the single edge joining z to the root o_{X_i} of X_i . Applying Lemma 3.1 to the positive definite block diag(D_1, D_2, D_3) shows that

$$r - \sum_{i=1}^3 b_i^\top D_i^{-1} b_i \geq 0.$$

The sign of each joining edge disappears from this expression, because its square is one. Thus, by the definition of the root response,

$$b_i^\top D_i^{-1} b_i = e_{o_{X_i}}^\top D_i^{-1} e_{o_{X_i}} = g_\varepsilon(X_i, \sigma).$$

We therefore obtain

$$r - \sum_{i=1}^3 g_\varepsilon(X_i, \sigma) \geq 0 \quad \text{for each } \varepsilon \in \{-1, 1\}.$$

Adding these two inequalities and using $s(X_i, \sigma) = g_1(X_i, \sigma) + g_{-1}(X_i, \sigma)$ gives

$$0 \leq 2r - \sum_{i=1}^3 s(X_i, \sigma).$$

However, (7) at $j = 61$ gives $s(X_i, \sigma) > c_{61} = 4\sqrt{2}/3$ for each $i = 1, 2, 3$. Hence

$$0 \leq 2r - \sum_{i=1}^3 s(X_i, \sigma) < 4\sqrt{2} - 3 \left(\frac{4\sqrt{2}}{3} \right) = 0,$$

a contradiction. Thus every signing of J has norm strictly greater than r . We arrive at the conclusion.

It remains to prove (7). We prove (7) by induction on j . Lemma 3.4 proves the case $j = 0$ since $Q_0 = H_{62}$. Suppose it holds at level $j < 61$, and let X, Y be the two child branches of an

occurrence Z of Q_{j+1} . The two restrictions of σ can be different, but each satisfies the claimed bound. Lemma 3.3 gives

$$s(Z, \sigma) \geq \frac{4}{2r - s(X, \sigma) - s(Y, \sigma)} > \frac{2}{r - c_j}.$$

Lemma 3.3 gives $2r - s(X, \sigma) - s(Y, \sigma) > 0$. The other denominator is also positive, since $0 \leq j < 61$ implies $64 - j \geq 4$, and hence

$$r - c_j = \sqrt{2} \left(1 - \frac{1}{64 - j} \right) \geq \frac{3\sqrt{2}}{4} > 0.$$

Direct simplification gives

$$(8) \quad \frac{2}{r - c_j} = \sqrt{2} \frac{64 - j}{63 - j} = \sqrt{2} \left(1 + \frac{1}{63 - j} \right) = c_{j+1}.$$

This proves (7) for every required level. □

5. PROOFS OF LEMMAS 3.2 AND 3.3

In this section, we prove Lemmas 3.2 and 3.3. These results establish the positive definiteness of the branch matrices and the response inequality for joining two rooted graphs, which are the key ingredients in the proof of the main theorem.

Proof of Lemma 3.2. The key point is that every branch has an edge joining its root to a parent vertex outside the branch. This edge allows us to deduce strict positivity for the branch matrix from positive semidefiniteness of the matrix on J .

Fix $\varepsilon \in \{-1, 1\}$. The assumption $\|A_\sigma(J)\| \leq r = 2\sqrt{2}$ implies

$$rI_J - \varepsilon A_\sigma(J) \succeq 0.$$

We prove by induction on j that $rI_X - \varepsilon A_\sigma(X) \succ 0$ for every rooted copy X of Q_j appearing in J .

For $j = 0$, each such copy is a seed H_{62} , so the assertion follows from Lemma 3.4.

Suppose the assertion holds at level j , where $0 \leq j < 61$, and let X be a rooted copy of Q_{j+1} in J . By construction, $X = B(X_1, X_2)$, where X_1 and X_2 are vertex-disjoint rooted copies of Q_j . Denote their roots by o_1 and o_2 , and denote the new root of X by w . The induction hypothesis gives

$$D_i := rI_{X_i} - \varepsilon A_\sigma(X_i) \succ 0, \quad i = 1, 2.$$

Let v be the unique neighbor of w outside X . Such a vertex exists for every branch under consideration: it is the next join vertex toward the centre of J , or the centre z itself when X is a copy of Q_{61} . Moreover, v has no neighbors in X_1 or X_2 . The relevant connections are illustrated in Figure 3.

For $i = 1, 2$, define the column vector

$$b_i = -\varepsilon \sigma(\{w, o_i\}) e_{o_i} \in \mathbb{R}^{V(X_i)},$$

where e_{o_i} is the coordinate vector at the root o_i . Order the vertices in $V(X) \cup \{v\}$ by listing those of X_1 , those of X_2 , and then w, v . The corresponding principal submatrix of $rI_J - \varepsilon A_\sigma(J)$ is

$$\begin{pmatrix} D_1 & 0 & b_1 & 0 \\ 0 & D_2 & b_2 & 0 \\ b_1^\top & b_2^\top & r & a \\ 0 & 0 & a & r \end{pmatrix} \succeq 0,$$

where $a := -\varepsilon\sigma(\{w, v\})$. Since $D_1, D_2 \succ 0$, Lemma 3.1 allows us to eliminate these two blocks. The resulting Schur complement is

$$\begin{pmatrix} p & a \\ a & r \end{pmatrix} \succeq 0, \quad p := r - b_1^\top D_1^{-1} b_1 - b_2^\top D_2^{-1} b_2.$$

Its determinant must be nonnegative. Therefore

$$pr - a^2 = pr - 1 \geq 0,$$

which implies $p \geq \frac{1}{r} > 0$.

Now consider the matrix on $X = B(X_1, X_2)$ alone:

$$rI_X - \varepsilon A_\sigma(X) = \begin{pmatrix} D_1 & 0 & b_1 \\ 0 & D_2 & b_2 \\ b_1^\top & b_2^\top & r \end{pmatrix}.$$

The Schur complement of $\text{diag}(D_1, D_2)$ is the same scalar p . Since D_1 and D_2 are positive definite and $p > 0$, Lemma 3.1 gives

$$rI_X - \varepsilon A_\sigma(X) \succ 0.$$

This completes the induction.

The argument applies separately to $\varepsilon = 1$ and $\varepsilon = -1$, with the same signing σ . Thus both branch matrices are positive definite for every required copy X , and all the stated root responses are well defined. $\square$

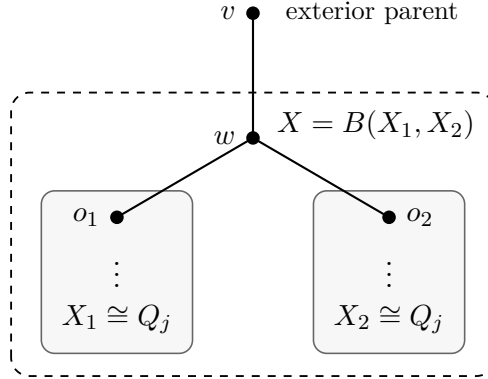


FIGURE 3. The branch $X = B(X_1, X_2)$ and its exterior parent v . The child branches X_1 and X_2 are vertex-disjoint copies of Q_j , rooted at o_1 and o_2 , respectively. The new root w is joined to o_1 , o_2 , and v . There are no edges between X_1 and X_2 , and v has no neighbors in either child branch. The dashed boundary encloses X ; adjoining v gives the vertex set indexing the principal submatrix used in the proof. When X is a copy of Q_{61} , its exterior parent is $v = z$. The internal structures of the child branches are omitted.

Proof of Lemma 3.3. We first compute the response at the new root by block elimination, and then combine the formulas for the two spectral sides.

Let o_X and o_Y be the roots of X and Y , respectively, and let o_Z be the new root of $Z = B(X, Y)$. Recall that $B(X, Y)$ is obtained from the disjoint union of X and Y by adding a new vertex o_Z and the two edges $\{o_Z, o_X\}$ and $\{o_Z, o_Y\}$, with o_Z designated as its root. Denote the signs of the two joining edges by

$$\tau_X = \sigma(\{o_Z, o_X\}), \quad \tau_Y = \sigma(\{o_Z, o_Y\}).$$

Thus $\tau_X, \tau_Y \in \{-1, 1\}$.

Fix $\varepsilon \in \{-1, 1\}$ and set

$$D_X = rI_X - \varepsilon A_\sigma(X), \quad D_Y = rI_Y - \varepsilon A_\sigma(Y).$$

Both matrices are positive definite, since they are principal submatrices of the positive definite matrix $rI_Z - \varepsilon A_\sigma(Z) \succ 0$. In particular, their inverses exist and the corresponding root responses g_ε are well defined.

Order the vertices of Z by listing those of X first, those of Y second, and o_Z last. Since the only edges between these three parts are the two joining edges, we have

$$rI_Z - \varepsilon A_\sigma(Z) = \begin{pmatrix} D_X & 0 & -\varepsilon \tau_X e_{o_X} \\ 0 & D_Y & -\varepsilon \tau_Y e_{o_Y} \\ -\varepsilon \tau_X e_{o_X}^\top & -\varepsilon \tau_Y e_{o_Y}^\top & r \end{pmatrix}.$$

Here e_{o_X} and e_{o_Y} are the coordinate vectors at the respective roots.

Apply Lemma 3.1 to eliminate the positive definite block $\text{diag}(D_X, D_Y)$. Its Schur complement is the scalar

$$\begin{aligned} r - (-\varepsilon \tau_X)^2 e_{o_X}^\top D_X^{-1} e_{o_X} - (-\varepsilon \tau_Y)^2 e_{o_Y}^\top D_Y^{-1} e_{o_Y} \\ = r - g_\varepsilon(X, \sigma) - g_\varepsilon(Y, \sigma) = p_\varepsilon > 0, \end{aligned}$$

where we used $\varepsilon^2 = \tau_X^2 = \tau_Y^2 = 1$. Since the full matrix is positive definite, Lemma 3.1 implies that its Schur complement $p_\varepsilon > 0$. Its inverse formula also shows that the diagonal entry of the inverse at o_Z is p_ε^{-1} . Hence

$$(9) \quad g_\varepsilon(Z, \sigma) = e_{o_Z}^\top (rI_Z - \varepsilon A_\sigma(Z))^{-1} e_{o_Z} = \frac{1}{p_\varepsilon}.$$

The argument applies to both $\varepsilon = 1$ and $\varepsilon = -1$, using the same signing σ . Adding the definitions of p_1 and p_{-1} gives

$$(10) \quad p_1 + p_{-1} = 2r - s(X, \sigma) - s(Y, \sigma) > 0.$$

For these two positive numbers,

$$(11) \quad \frac{1}{p_1} + \frac{1}{p_{-1}} - \frac{4}{p_1 + p_{-1}} = \frac{(p_1 - p_{-1})^2}{p_1 p_{-1} (p_1 + p_{-1})} \geq 0.$$

Combining (9), (10) and (11), we have

$$\begin{aligned} s(Z, \sigma) &= g_1(Z, \sigma) + g_{-1}(Z, \sigma) = \frac{1}{p_1} + \frac{1}{p_{-1}} \\ &\geq \frac{4}{p_1 + p_{-1}} = \frac{4}{2r - s(X, \sigma) - s(Y, \sigma)}. \end{aligned}$$

This proves the response formula, the stated inequality, and the positivity of its denominator. $\square$

6. PROOF OF LEMMA 3.4

We begin with an auxiliary lemma that provides the spectral bounds needed in the proof of Lemma 3.4. These bounds follow from known results for unsigned trees and unicyclic graphs [13, 6], but we include an elementary proof for completeness. We then compute the root response of the complete binary tree explicitly and use this calculation to prove Lemma 3.4.

Lemma 6.1. *Recall that $r := 2\sqrt{2}$. The following statements hold.*

- (i) *Let U be a finite connected graph with exactly one cycle and maximum degree at most three. Then every signing σ of U satisfies*

$$\|A_\sigma(U)\| < r.$$

- (ii) Let h be a nonnegative integer, and let T_h be the rooted binary tree defined in Section 2.1. For every signing σ of T_h and every $\varepsilon \in \{-1, 1\}$, we have

$$rI_{T_h} - \varepsilon A_\sigma(T_h) \succ 0.$$

Moreover, the corresponding root response is

$$(12) \quad g_\varepsilon(T_h, \sigma) = \frac{h+1}{(h+2)\sqrt{2}}, \quad \varepsilon \in \{-1, 1\}.$$

Proof. (i) *Graphs with exactly one cycle.* Let σ be an arbitrary signing of U . Write the vertices of its unique cycle in cyclic order as $c_0, \dots, c_{m-1}$, where $m \geq 3$. Orient the cycle edges from c_i to c_{i+1} , with indices taken modulo m .

Every component outside the cycle is a tree attached to exactly one cycle vertex: a cycle in such a component, or two distinct attachments to the original cycle, would create another cycle in U . Orient all edges of these attached trees away from the cycle. Thus every vertex has exactly one incoming edge. Since the maximum degree is at most three, its outdegree is at most two. Denote the outdegree of a vertex v by $d^+(v)$. An example of this orientation is shown in Figure 4(a).

Let $x = (x_v)_{v \in V(U)}$ be a real vector. For every directed edge $u \rightarrow v$, we have

$$(13) \quad 2|x_u x_v| \leq \frac{x_u^2}{\sqrt{2}} + \sqrt{2} x_v^2.$$

Equality holds precisely when $|x_u| = \sqrt{2}|x_v|$.

We now sum (13) over all edges. Each vertex contributes $\sqrt{2}x_v^2$ through its unique incoming edge and $d^+(v)x_v^2/\sqrt{2}$ through its outgoing edges. Therefore

$$(14) \quad \begin{aligned} |x^\top A_\sigma(U)x| &= \left| 2 \sum_{\{u,v\} \in E(U)} \sigma(\{u,v\}) x_u x_v \right| \\ &\leq 2 \sum_{\{u,v\} \in E(U)} |x_u x_v| \\ &\leq \sum_{v \in V(U)} \left(\sqrt{2} + \frac{d^+(v)}{\sqrt{2}} \right) x_v^2 \\ &\leq r \|x\|_2^2. \end{aligned}$$

The last inequality follows from $d^+(v) \leq 2$.

To obtain a strict bound, first observe that at least one vertex has outdegree less than two. Indeed, deleting one cycle edge leaves a tree on the same vertex set, so $|E(U)| = |V(U)|$. Hence

$$\sum_{v \in V(U)} d^+(v) = |E(U)| = |V(U)|,$$

which rules out $d^+(v) = 2$ at every vertex.

Suppose that equality holds in (14) for some nonzero vector x . Then equality must hold at every step of that chain. In particular,

$$\sum_{v \in V(U)} \frac{2 - d^+(v)}{\sqrt{2}} x_v^2 = 0.$$

Every summand is nonnegative. Thus $x_v = 0$ at every vertex whose outdegree is less than two, and at least one such vertex exists.

Equality must also hold in (13) on every edge, since the sum of its nonnegative differences is zero. Consequently,

$$|x_u| = \sqrt{2}|x_v| \quad \text{whenever } u \rightarrow v.$$

If either endpoint of an edge has zero coordinate, this identity forces the other endpoint to have zero coordinate as well. Since U is connected, the zero coordinate found above propagates along paths to every vertex. This gives $x = 0$, a contradiction.

We have therefore proved

$$|x^\top A_\sigma(U)x| < r\|x\|_2^2 \quad \text{for every nonzero } x.$$

The function $x \mapsto |x^\top A_\sigma(U)x|$ attains its maximum on the unit sphere, and the spectral theorem identifies this maximum with $\|A_\sigma(U)\|$. It follows that $\|A_\sigma(U)\| < r$.

(ii) *Full rooted binary trees.* We prove the positive definiteness and the response formula simultaneously by induction on the height h .

When $h = 0$, the tree T_0 consists of its root alone. Its adjacency matrix is zero, so for either $\varepsilon \in \{-1, 1\}$,

$$rI_{T_0} - \varepsilon A_\sigma(T_0) = (r) \succ 0, \quad g_\varepsilon(T_0, \sigma) = \frac{1}{r} = \frac{1}{2\sqrt{2}}.$$

This is (12) at $h = 0$.

Suppose the assertions hold at height h , and let σ be any signing of T_{h+1} . Denote its root by o . Removing o leaves two vertex-disjoint rooted copies X_1, X_2 of T_h , whose roots are denoted by o_1, o_2 . Thus $T_{h+1} = B(X_1, X_2)$, as illustrated in Figure 4(b). We also write σ for the restriction of the signing to either child tree.

Fix $\varepsilon \in \{-1, 1\}$ and define

$$D_i := rI_{X_i} - \varepsilon A_\sigma(X_i), \quad i = 1, 2.$$

By the induction hypothesis, both matrices are positive definite and

$$e_{o_i}^\top D_i^{-1} e_{o_i} = g_\varepsilon(X_i, \sigma) = \frac{h+1}{(h+2)\sqrt{2}}, \quad i = 1, 2,$$

where e_{o_i} is the coordinate vector at o_i . The restrictions of σ to the two child trees may differ; the induction hypothesis applies to each of them.

Let

$$\tau_i := \sigma(\{o, o_i\}), \quad b_i := -\varepsilon \tau_i e_{o_i}, \quad i = 1, 2.$$

Ordering the vertices of X_1 first, those of X_2 second, and o last gives

$$rI_{T_{h+1}} - \varepsilon A_\sigma(T_{h+1}) = \begin{pmatrix} D_1 & 0 & b_1 \\ 0 & D_2 & b_2 \\ b_1^\top & b_2^\top & r \end{pmatrix}.$$

The Schur complement of $\text{diag}(D_1, D_2)$ is the scalar

$$\begin{aligned} p_\varepsilon &:= r - b_1^\top D_1^{-1} b_1 - b_2^\top D_2^{-1} b_2 \\ &= r - g_\varepsilon(X_1, \sigma) - g_\varepsilon(X_2, \sigma) \\ &= 2\sqrt{2} - \frac{2(h+1)}{(h+2)\sqrt{2}} \\ &= \frac{(h+3)\sqrt{2}}{h+2} > 0. \end{aligned}$$

The second equality uses $(-\varepsilon \tau_i)^2 = 1$ for $i = 1, 2$.

Since $D_1, D_2 \succ 0$ and $p_\varepsilon > 0$, Lemma 3.1 proves that

$$rI_{T_{h+1}} - \varepsilon A_\sigma(T_{h+1}) \succ 0.$$

The inverse formula in the same lemma gives the diagonal entry of the inverse at the root:

$$g_\varepsilon(T_{h+1}, \sigma) = \frac{1}{p_\varepsilon} = \frac{h+2}{(h+3)\sqrt{2}}.$$

This is the required formula at height $h + 1$. Since σ and ε were arbitrary, the induction proves both assertions for every height, every signing, and both spectral sides. $\square$

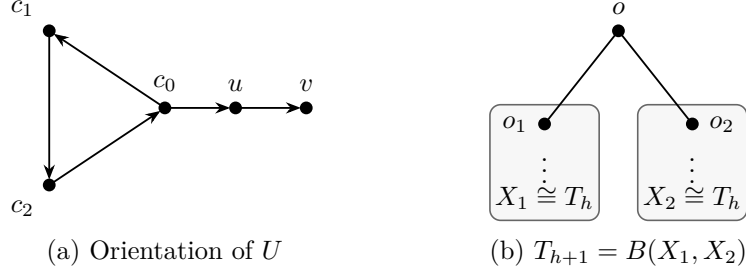


FIGURE 4. The two structures used in the proof. (a) An example with a three-vertex cycle $c_0c_1c_2c_0$ and an attached path c_0uv . The cycle is oriented cyclically, and the attached path is oriented away from it. (b) The root o of T_{h+1} is joined to the roots o_1, o_2 of two disjoint copies X_1, X_2 of T_h . The shaded regions represent the child trees, whose internal structures are omitted. For $h = 0$, each child tree consists only of its root.

We next present the proof of Lemma 3.4.

Proof of Lemma 3.4. The graph H_{62} is connected, has exactly one cycle, and has maximum degree three. By Lemma 6.1, for every signing σ of H_{62} , both matrices

$$rI_{H_{62}} - A_\sigma(H_{62}) \quad \text{and} \quad rI_{H_{62}} + A_\sigma(H_{62})$$

are positive definite.

Reducing the signs. For notational convenience, write $S := A_\sigma(H_{62})$ throughout this proof. We simplify the edge signs by *vertex switching*. Specifically, choose an auxiliary sign $\delta_u \in \{-1, 1\}$ at each vertex u and replace the sign of each edge uv by $\delta_u \sigma(\{u, v\}) \delta_v$. The resulting signed adjacency matrix is $\Delta S \Delta$, where $\Delta = \text{diag}(\delta_u)_{u \in V(H_{62})}$. Since $\Delta^{-1} = \Delta$, this transformation preserves the spectrum. It also preserves the diagonal entries of the inverses appearing in (4). Indeed, whenever $rI_{H_{62}} - \varepsilon S$ is invertible,

$$(rI_{H_{62}} - \varepsilon \Delta S \Delta)^{-1} = \Delta (rI_{H_{62}} - \varepsilon S)^{-1} \Delta.$$

Conjugation by Δ leaves each diagonal entry unchanged, because $\delta_u^2 = 1$ for every vertex u . In particular, vertex switching preserves the root response $g_\varepsilon(H_{62}, \sigma)$.

Deleting the edge $\{v_1, v_2\}$ from H_{62} yields a tree on the same vertex set. We choose Δ so that every edge of the spanning tree has sign $+1$ after switching. Write δ_u for the diagonal entry of Δ at a vertex u , and set $\delta_o = 1$. Proceeding from the root toward the leaves, define $\delta_v = \delta_u \sigma(\{u, v\})$ whenever u is the parent of v in the tree. Then the transformed sign on this edge is $\delta_u \sigma(\{u, v\}) \delta_v = 1$. This determines Δ on all vertices.

In the switched signed adjacency matrix $\Delta S \Delta$, every edge except possibly $\{v_1, v_2\}$ has sign $+1$. The sign on $\{v_1, v_2\}$ is

$$\tau := \sigma(\{v_0, v_1\}) \sigma(\{v_1, v_2\}) \sigma(\{v_2, v_0\}) \in \{-1, 1\}.$$

Indeed, replacing S by $\Delta S \Delta$ preserves the product of the edge signs around the triangle: each diagonal entry of Δ associated with a triangle vertex appears twice in this product and therefore contributes a factor of 1. Since the entries of $\Delta S \Delta$ corresponding to $\{v_0, v_1\}$ and $\{v_0, v_2\}$ are both $+1$, its entries corresponding to $\{v_1, v_2\}$ must equal τ .

Eliminating the trees. Work first with $rI_{H_{62}} - \Delta S \Delta$ after this sign change. Each of the three attached trees has response $63/(64\sqrt{2})$ by (12). In this calculation set

$$a = r - \frac{63}{64\sqrt{2}} = \frac{193}{64\sqrt{2}}.$$

In the matrix $rI_{H_{62}} - \Delta S \Delta$, take the Schur complement of the block diagonal principal submatrix indexed by the vertices of the three attached copies of T_{62} . This eliminates those tree vertices and retains only o, v_0, v_1, v_2 . We use M_τ to denote the Schur complement. Before eliminating the trees, the principal submatrix corresponding to o, v_0, v_1, v_2 , in this order, is

$$\begin{pmatrix} r & -1 & 0 & 0 \\ -1 & r & -1 & -1 \\ 0 & -1 & r & -\tau \\ 0 & -1 & -\tau & r \end{pmatrix} \begin{matrix} o \\ v_0 \\ v_1 \\ v_2 \end{matrix}.$$

Since the three trees are disjoint and each is attached to only one retained vertex, the correction term in the Schur complement is

$$\text{diag}\left(\frac{63}{64\sqrt{2}}, 0, \frac{63}{64\sqrt{2}}, \frac{63}{64\sqrt{2}}\right).$$

Subtracting this term replaces the diagonal entries at o, v_1, v_2 by a , while leaving the diagonal entry at v_0 and all off-diagonal entries unchanged. This gives

$$(15) \quad M_\tau = \begin{pmatrix} a & -1 & 0 & 0 \\ -1 & r & -1 & -1 \\ 0 & -1 & a & -\tau \\ 0 & -1 & -\tau & a \end{pmatrix} \begin{matrix} o \\ v_0 \\ v_1 \\ v_2 \end{matrix}.$$

For each $\tau \in \{-1, 1\}$, the matrix M_τ arises from a signing of the seed and is therefore positive definite. Let u_τ denote the diagonal entry of M_τ^{-1} corresponding to o . By the inverse-block formula in Lemma 3.1 and the invariance of the root response under switching,

$$g_1(H_{62}, \sigma) = u_\tau = (M_\tau^{-1})_{11}.$$

A direct calculation using the cofactor formula gives

$$(16) \quad u_\tau = \frac{ra - 2 - r\tau}{a^2r - 3a + (1 - ar)\tau}.$$

Adding the two spectral sides. Negating the original signing negates the product around its three-edge cycle. After the same reduction, the negative signing therefore has parameter $-\tau$. Consequently the two responses of the original signing are u_1, u_{-1} , in some order, and $s(H_{62}, \sigma) = u_\tau + u_{-\tau} = u_1 + u_{-1}$ is independent of the signing.

Substitution of $a = 193/(64\sqrt{2})$ and $r = 2\sqrt{2}$ into (16) gives the particularly short expression

$$\frac{u_\tau}{\sqrt{2}} = \frac{8256 - 4096\sqrt{2}\tau}{18721 - 10304\sqrt{2}\tau}.$$

For clarity, the full addition is

$$(17) \quad \begin{aligned} \frac{s(H_{62}, \sigma)}{\sqrt{2}} &= \frac{8256 - 4096\sqrt{2}}{18721 - 10304\sqrt{2}} + \frac{8256 + 4096\sqrt{2}}{18721 + 10304\sqrt{2}} \\ &= \frac{2(8256 \cdot 18721 - 2 \cdot 4096 \cdot 10304)}{18721^2 - 2 \cdot 10304^2} = \frac{140300416}{138131009} > \frac{65}{64}. \end{aligned}$$

□

7. WHY THIS BASE GRAPH F IS NOT RAMANUJAN

In this section, we show that the constructed graph F is not Ramanujan by proving that $\lambda_2(A(F)) > 2\sqrt{2}$. We choose two disjoint completed seeds of equal order with no edge between them and define a vector taking opposite constant values on these subgraphs and zero elsewhere. This vector is orthogonal to the constant vector and has Rayleigh quotient greater than $2\sqrt{2}$. The variational characterization of $\lambda_2(A(F))$ then gives the desired conclusion.

Proposition 7.1. *Let $\lambda_2(A(F))$ denote the second largest eigenvalue of $A(F)$, with eigenvalues ordered nonincreasingly and counted with multiplicity. Then*

$$\lambda_2(A(F)) > 2\sqrt{2}.$$

In particular, F is not Ramanujan.

Proof. Let C be the induced subgraph consisting of one copy of H_{62} and the completion vertices attached to its leaf triples. By construction, each such triple lies entirely within this seed. Hence

$$n_C := |V(C)| = (3 \cdot 2^{63} + 1) + 2 \left(\frac{3 \cdot 2^{62}}{3} \right) = 2^{65} + 1.$$

Writing o_C for the seed root, we have

$$\deg_C(v) = \begin{cases} 2, & v = o_C, \\ 3, & v \in V(C) \setminus \{o_C\}. \end{cases}$$

The degree-sum formula therefore gives

$$(18) \quad 2|E(C)| = \sum_{v \in V(C)} \deg_C(v) = 3n_C - 1.$$

Choose the two seed copies joined at the root w of one copy of $Q_1 = B(H_{62}, H_{62})$. Let C_1, C_2 be their completed seeds, with roots o_1, o_2 , respectively. The vertex w belongs to neither completed seed, and

$$V(C_1) \cap V(C_2) = \emptyset, \quad |V(C_1)| = |V(C_2)| = n_C.$$

There is no edge between C_1 and C_2 . For each $i = 1, 2$, the only edge from C_i to its complement in F is $\{o_i, w\}$; see Figure 5.

Define $x \in \mathbb{R}^{V(F)}$ by

$$x_v = \begin{cases} (2n_C)^{-1/2}, & v \in V(C_1), \\ -(2n_C)^{-1/2}, & v \in V(C_2), \\ 0, & v \notin V(C_1) \cup V(C_2). \end{cases}$$

If $\mathbf{1}$ denotes the vector with every coordinate equal to one, then

$$\|x\|_2^2 = \frac{n_C + n_C}{2n_C} = 1, \quad x^\top \mathbf{1} = \frac{n_C - n_C}{\sqrt{2n_C}} = 0.$$

Edges leaving either completed seed contribute zero to $x^\top A(F)x$, because their other endpoint has coordinate zero. Consequently,

$$\begin{aligned} x^\top A(F)x &= 2 \sum_{\{u,v\} \in E(F)} x_u x_v \\ &= \frac{|E(C_1)| + |E(C_2)|}{n_C} \\ &= \frac{3n_C - 1}{n_C} = 3 - \frac{1}{n_C}, \end{aligned}$$

where the third equality follows from (18).

To apply the variational characterization of λ_2 , recall that F is connected and cubic. Thus

$$A(F)\mathbf{1} = 3 \cdot \mathbf{1},$$

and, for every $y \in \mathbb{R}^{V(F)}$,

$$y^\top (3I_F - A(F))y = \sum_{\{u,v\} \in E(F)} (y_u - y_v)^2 \geq 0.$$

Connectedness implies that equality holds precisely when y is constant. Hence 3 is the largest eigenvalue of $A(F)$ and its eigenspace is $\text{span}\{\mathbf{1}\}$. The spectral theorem therefore gives

$$\begin{aligned} \lambda_2(A(F)) &= \max_{\substack{y \in \mathbb{R}^{V(F)} \\ \|y\|_2=1, y^\top \mathbf{1}=0}} y^\top A(F)y \\ &\geq x^\top A(F)x \\ &= 3 - \frac{1}{2^{65} + 1}. \end{aligned}$$

Finally, $n_C > 6$ and $3 - \frac{1}{n_C} > \frac{17}{6} > 2\sqrt{2}$. Since $\lambda_2(A(F)) < 3$ and $\lambda_2(A(F)) > 2\sqrt{2}$, it is a nontrivial eigenvalue outside the Ramanujan interval. Thus F is not Ramanujan. $\square$

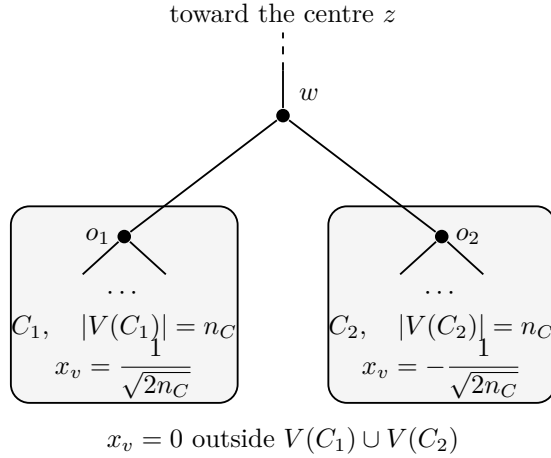


FIGURE 5. Two completed seeds C_1, C_2 with a common exterior parent w . Each box represents an entire induced subgraph, with its internal structure omitted. The only edge leaving C_i is $\{o_i, w\}$, and there is no edge between C_1 and C_2 . The displayed values define the test vector used to estimate $\lambda_2(A(F))$.

8. CONCLUSION

We have constructed a finite connected simple regular graph F for which every signing has an eigenvalue outside the Ramanujan interval, disproving the Bilu–Linial signing conjecture for general regular graphs. The graph F is not Ramanujan, so this construction does not settle the conjecture restricted to Ramanujan base graphs.

Two questions remain for further study. First, does every finite connected simple regular Ramanujan graph admit a signing whose entire spectrum lies in the Ramanujan interval? Second, does every finite connected simple regular graph admit an ℓ -lift, for some integer $\ell \geq 2$, whose new eigenvalues all lie in the Ramanujan interval? Here the interval is determined by the degree of the base graph, and ℓ may depend on the base graph. The second question asks whether allowing more sheets can overcome the obstruction to the desired bound for 2-lifts.

Statement on the Use of AI. The counterexample and its proof strategy were developed with the assistance of ChatGPT. The author has carefully reviewed all AI-generated text and verified the mathematical arguments, revising or rewriting the material as necessary. The author takes full responsibility for the content and correctness of this paper.

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